

Network architecture of the long-distance pathways in the macaque brain

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Understanding the network structure of white matter communication pathways is essential for unraveling the mysteries of the brain's function, organization, and evolution. To this end, we derive a unique network incorporating 410 anatomical tracing studies of the macaque brain from the Collation of Connectivity data on the Macaque brain (CoCoMac) neuroinformatic database. Our network consists of 383 hierarchically organized regions spanning cortex, thalamus, and basal ganglia; models the presence of 6,602 directed long-distance connections; is three times larger than any previously derived brain network; and contains subnetworks corresponding to classic corticocortical, corticosubcortical, and subcortico-subcortical fiber systems. We found that the empirical degree distribution of the network is consistent with the hypothesis of the maximum entropy exponential distribution and discovered two remarkable bridges between the brain's structure and function via network-theoretical analysis. First, prefrontal cortex contains a disproportionate share of topologically central regions. Second, there exists a tightly integrated core circuit, spanning parts of premotor cortex, prefrontal cortex, temporal lobe, parietal lobe, thalamus, basal ganglia, cingulate cortex, insula, and visual cortex, that includes much of the task-positive and task-negative networks and might play a special role in higher cognition and consciousness.

neuroanatomy | brain network | network analysis | structural | functional

In 1669, Nicolaus Steno (1) referred to white matter as “nature’s finest masterpiece.” White matter pathways in the brain mediate information flow and facilitate information integration and co-operation across functionally differentiated distributed centers of sensation, perception, action, cognition, and emotion. Uncovering the global topological regularities of the logical long-distance connections that are subserved by the physical white matter pathways is a key prerequisite to any theory of brain function, dysfunction, organization, dynamics, and evolution.

Anatomical tracing in experimental animals has historically been the pervasive technique for mapping long-distance white matter projections (2–4). Given the resolution of anatomical tracing experiments, they typically furnish data at a macroscale of cortical areas or, more generally, brain regions. The associated network description* models brain regions as vertices and the presence of reported long-distance connections as directed edges between them.

The most well-known network of the macaque monkey visual cortex consists of 32 vertices and 305 edges (2). Other networks of the macaque cortex consist of 70 vertices and 700 edges (5) and 95 vertices and 2,402 edges (6). The largest network of the cat cortex has 95 vertices and 1,500 edges (7). Network-theoretical analyses have uncovered a number of remarkable insights: distributed and hierarchical structure of cortex (2); topological organization of cortex (8); indeterminacy of unique hierarchy (9); functional small-world characteristics, optimal set analysis, and multidimensional scaling (10); small-world characteristics (11); nonoptimal component placement for wire length (6); structural and functional motifs (12); and hub identification and classification (13). However, even the largest previous network (6) completely lacks edges corresponding to corticosubcortical and subcortico-subcortical long-distance connections and has significant gaps even among corticocortical long-distance connections (*SI Appendix, Fig. S1*).

To gain a better understanding of the structure and organization of the brain, a network spanning the entire brain would be extremely useful. Such a network will be an indispensable foundation for clinical, systems, cognitive, and computational neurosciences (14). No such network has been reported. We undertake the challenge of constructing, visualizing, and analyzing such a network. Our network opens the door to the application of large-scale network-theoretical analysis that has been so successful in understanding the Internet (15), metabolic networks, protein interaction networks (16), various social networks (17), and searching the World-Wide Web (18, 19).

Model: Deriving the Network Description

Collation of Connectivity data on the Macaque brain (CoCoMac), a seminal contribution to neuroinformatics, is a publicly available database (20–22). Conscientiously and meticulously, the database curators have collated and annotated information on over 2,500 anatomical tracer injections from over 400 published experimental studies.

CoCoMac is an objective, coordinate-independent collection of annotations that captures two relationships between pairs of brain regions, where each brain region refers to cortical and subcortical subdivisions as well as to combinations of such subdivisions into sulci, gyri, and other large ensembles. The first relationship is connectivity—whether a brain region in one study projects to another region in (possibly) a different study. There are 10,681 connectivity relations.[†] The second relationship is mapping—whether a brain region in one study is identical to, a substructure of, or a suprastructure of another region in (possibly) a different study. There are 16,712 mapping relations. Unfortunately, because of a multiplicity of brain maps, divergent nomenclature, boundary uncertainty, and differing resolutions in different studies, mapping relations are often conflicting and connectivity information is typically scattered across related brain regions. The situation is aptly described by Van Essen (23): “Our fragmentary and rapidly evolving understanding is reminiscent of the situation faced by cartographers of the earth’s surface many centuries ago, when maps were replete with uncertainties and divergent portrayals of most of the planet’s surface.” Consolidating connectivity information by merging logically equivalent brain regions and aggregating their connectivity is a necessary prerequisite to any network-analytical study. Further, it is desirable to place the merged brain regions into a coherent, unified, hierarchical brain map that recursively partitions brain and its constituents into

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*It is important to draw a distinction between the actual physical network in a macaque brain and its logical description in network-theoretical terminology using reported data. Because we are primarily concerned with the latter usage in this paper, we will refer to network description as network.

[†]CoCoMac also reports 13,498 plausible connections that were tested for but were not found. This substantially reduces the possibility that projections present in the brain are dramatically undersampled or underreported.

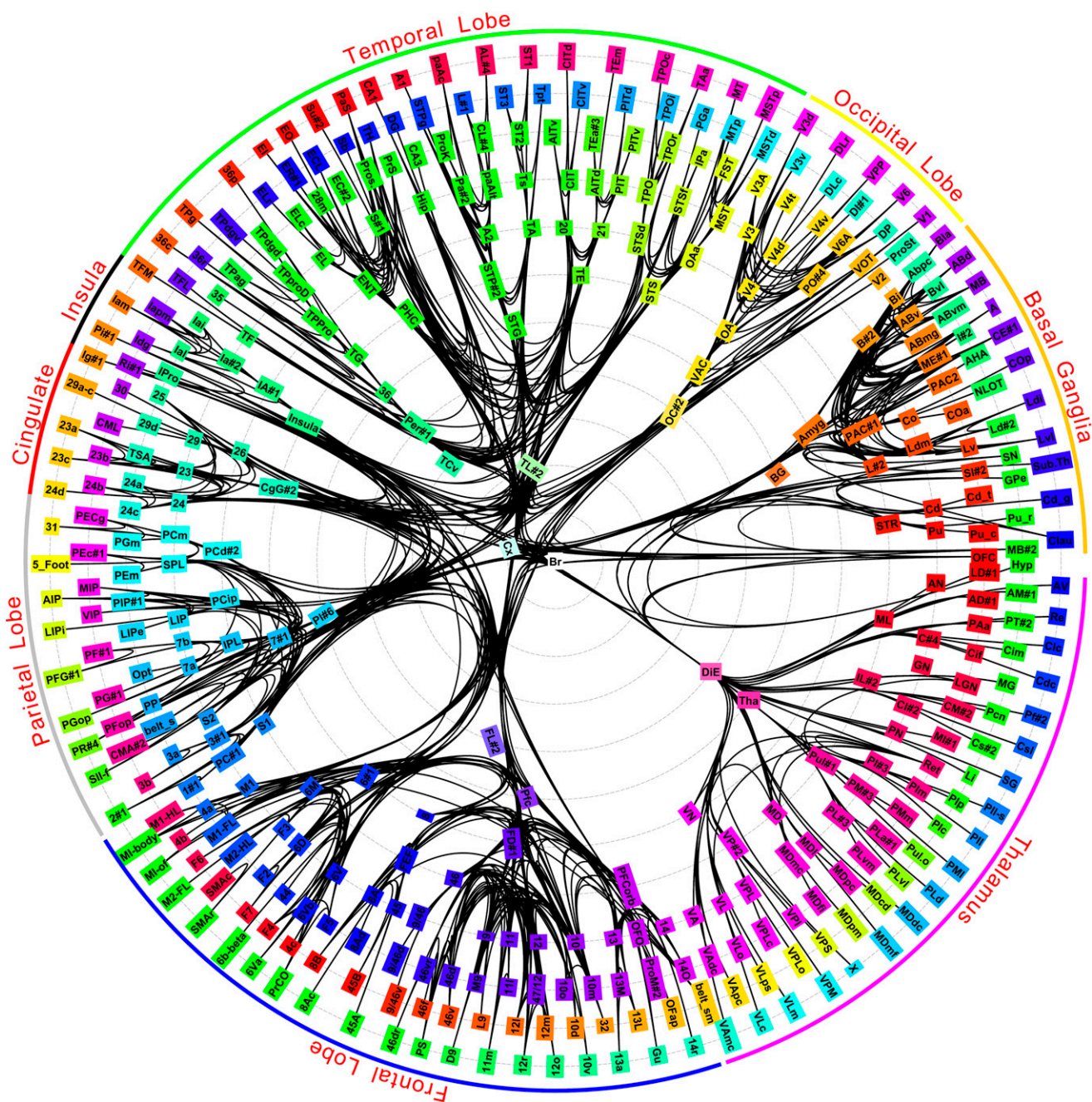


Fig. 1. Macaque brain long-distance network. Each vertex of the network corresponds to a brain region in the hierarchical brain map of *SI Appendix, Fig. S6*, and each edge encodes the presence of long-distance connection between corresponding brain regions. Edges are drawn using algorithmically bundled splines (25). *SI Appendix, Tables S2 and S3* provide a summary of the number of edges in major corticocortical and corticosubcortical subnetworks. A color wheel is used for better discrimination amongst brain regions. For the leaf brain regions in the two outermost circles, the color wheel is rotated by 120° and 240°. The edges are drawn in black. *SI Appendix, Table S1* enumerates the entire hierarchical brain map and provides a complete index to acronyms of the brain regions; it has been color-coded for wider accessibility.

progressively smaller physical regions.[‡] The brain map can provide a natural frame of reference within which to correlate, aggregate, and understand various merged brain regions. Conceptually, merging brain regions and extracting a hierarchy can be carried out according to logical and formal calculus developed by CoCoMac curators (20–22, 24). In practice, the tasks are made formidable by a number of factors. For example, (i) there are partially over-

lapping brain regions (*SI Appendix, Fig. S5*); (ii) there are direct conflicts between mapping relations (*SI Appendix, Fig. S3*); (iii) there are implied indirect conflicts that are far too numerous and inherently insidious (*SI Appendix, Fig. S3*); and (iv) there are errors and omissions in the underlying database, which itself is large. Although it is difficult to define a formal metric against which a single hierarchical brain map can be defensibly constructed, reassuringly, any hierarchical brain map built on the same set of merged regions will at most affect the resolution of the network-theoretical analysis. In this study, we have constructed one hierarchical brain map, at the highest resolution that the data can meaningfully support, toward our goal of network analysis

[‡]This usage of physical hierarchical partition of brain into its constituent parts is different from logical hierarchical information processing in visual cortex, as discussed in the article by Felleman and Van Essen (2).

(SI Appendix). The entire set of merged brain regions and our hierarchical brain map are explicitly detailed in the multipage SI Appendix, Table S1 to provide complete transparency and to permit future additions, deletions, and modifications as data with finer resolution become available.

SI Appendix, Fig. S6 visualizes our hierarchical brain map. It can be seen that the brain is divided into cortex, diencephalon, and basal ganglia, which are themselves divided into smaller regions, such as temporal lobe, frontal lobe, parietal lobe, occipital lobe, insula, and cingulate cortex. With the brain regions in the hierarchical brain map as vertices, our network contains 6,602 edges, wherein an edge encodes the presence of long-distance connection between corresponding brain regions. Fig. 1 displays the network on the hierarchical brain map, where each edge is visualized by a spline curve. Visualizing 6,602 edges directly leads to a highly cluttered figure in which no details are discernible (SI Appendix, Fig. S17A). To improve clarity, splines with a common origin or destination are bundled algorithmically (25) (SI Appendix and SI Appendix, Figs. S16 and S17). The figure succinctly captures many aspects of the cumulative contribution of a whole community of neuroanatomists over the past half century into a single illustration.

The long distance network dataset consists of three text files: *Macaque_LongDistance_Network.nameslist*, *Macaque_LongDistance_Network_connectivity.edgelist*, and *Macaque_LongDistance_Network_mapping.edgelist*. The files are publicly available and are described in SI Appendix.

Our network is (i) comprehensive in that it incorporates every study included in CoCoMac; (ii) consistent in that every edge can be tracked back to an underlying tracer study; (iii) concise in that identical brain regions (e.g., V1, 17, striate cortex) are merged and their connectivity is aggregated, thus reducing brain regions to 383 from 6,877 in the original database; (iv) coherent in that brain regions are organized in a unified hierarchical parcellation or brain map; and, finally, (v) colossal in that it is roughly three times larger than the largest previous such network (6) (compare Fig. 1 with SI Appendix, Fig. S1).

The comprehensiveness of our network is underscored by the fact that it contains logical subnetworks corresponding to a number of important physical fiber systems, namely, the visual system (2); dorsal-ventral pathways (3); thalamocortical relays (26); and numerous corticocortical, corticosubcortical, and subcortico-subcortical fiber systems (4). The brain regions involved in these fiber systems are enumerated in SI Appendix, Table S4, and the corresponding subnetworks are illustrated in SI Appendix, Figs. S18–21. It is important to note that strength, trajectory, and laminar source/target of projections are missing from our network, which only encodes the presence of connections.

Preliminary analysis (SI Appendix) confirms that the network is sparse, reciprocal, and small-world (27, 11) and reveals that the network has the proverbial six degrees of separation (28). As our main contributions, we first characterize the degree distribution, that is, the probability distribution of the number of connections that each brain region makes. Second, we study topologically central regions and subnetworks in the brain and, in the process, reveal two remarkable anatomical substrates of behavior via network-theory and web-searching algorithms.

Results

Degree Distribution of the Brain Network. In a network, degree of a vertex is the total number of edges that it touches. The tail behavior of the frequency distribution of degrees is a key signature of how connectivity is spread among vertices. A scale-free network follows a power law; that is, asymptotically, the probability that a vertex is connected with k other vertices is proportional to $k^{-\gamma}$ for some positive power γ . Scale-free networks naturally arise via mechanisms of growth and preferential attachment (29). For an exponential network, asymptotically, the probability that a vertex is connected with k other vertices is proportional to $e^{-k/\lambda}$, for some positive constant λ . Exponential networks can arise via random network evolution (30) or via a mechanism that hinders preferential attachment (31), such as

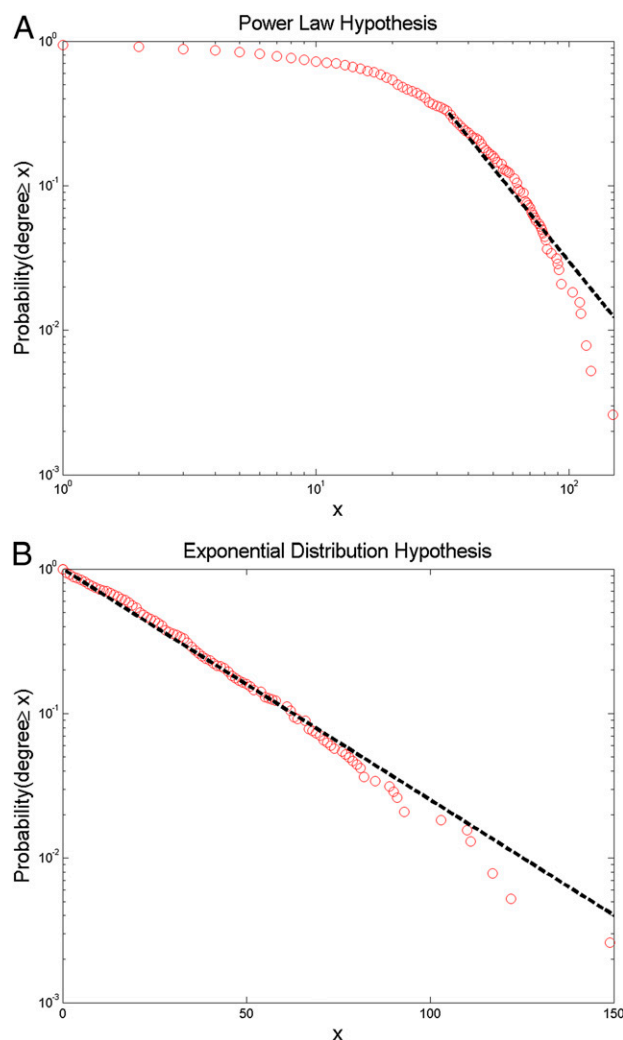


Fig. 2. Our network is directed, meaning that each edge is an ordered pair of vertices. By keeping the connectivity but removing direction, we created the undirected version of our network that has 383 vertices and 5,208 edges. The undirected network has an average degree of $\lambda = 27.2$. Following Keller (39), we analyze the behavior of the empirical complementary cumulative degree distribution (also known as survival function), which is drawn using circles on both of the above plots. The dashed line in the top log-log plot shows the complementary cumulative distribution of the maximum likelihood power law fit, $\sim x^{-3.15}$, $x \geq 33$, which was derived using the software provided with Clauset et al. (37). Moreover, the P value is extremely small ($\ll 0.1$); hence, the maximum likelihood power law hypothesis is rejected (37, box 1). The dashed line in the bottom log-linear plot shows the complementary cumulative distribution of the maximum entropy exponential distribution fit, $\lambda^{-1} \exp(-x/\lambda)$, over the entire range of data. The bottom plot is also shown using the linear-linear scale in SI Appendix, Fig. S22. These plots suggest that the hypothesis of the maximum entropy exponential distribution is consistent with the data.

the cost of adding links to the vertices or the limited capacity of a vertex. The World-Wide Web, the Internet (15), some social networks, and the metabolic networks are all scale-free (16), whereas power grids, air traffic networks, and collaboration networks of company directors (31, 16) are all exponential.

A simple but fundamental unanswered question is whether the degree distribution of the brain network is scale-free, exponential, or neither? In related work, Humphries et al. (32) reported that the brainstem reticular formation is not a scale-free network. For the smaller brain networks, Sporns and Zwi (11) did not find evidence for power law distribution but left open the possibility that a large-scale network may uncover such structure.

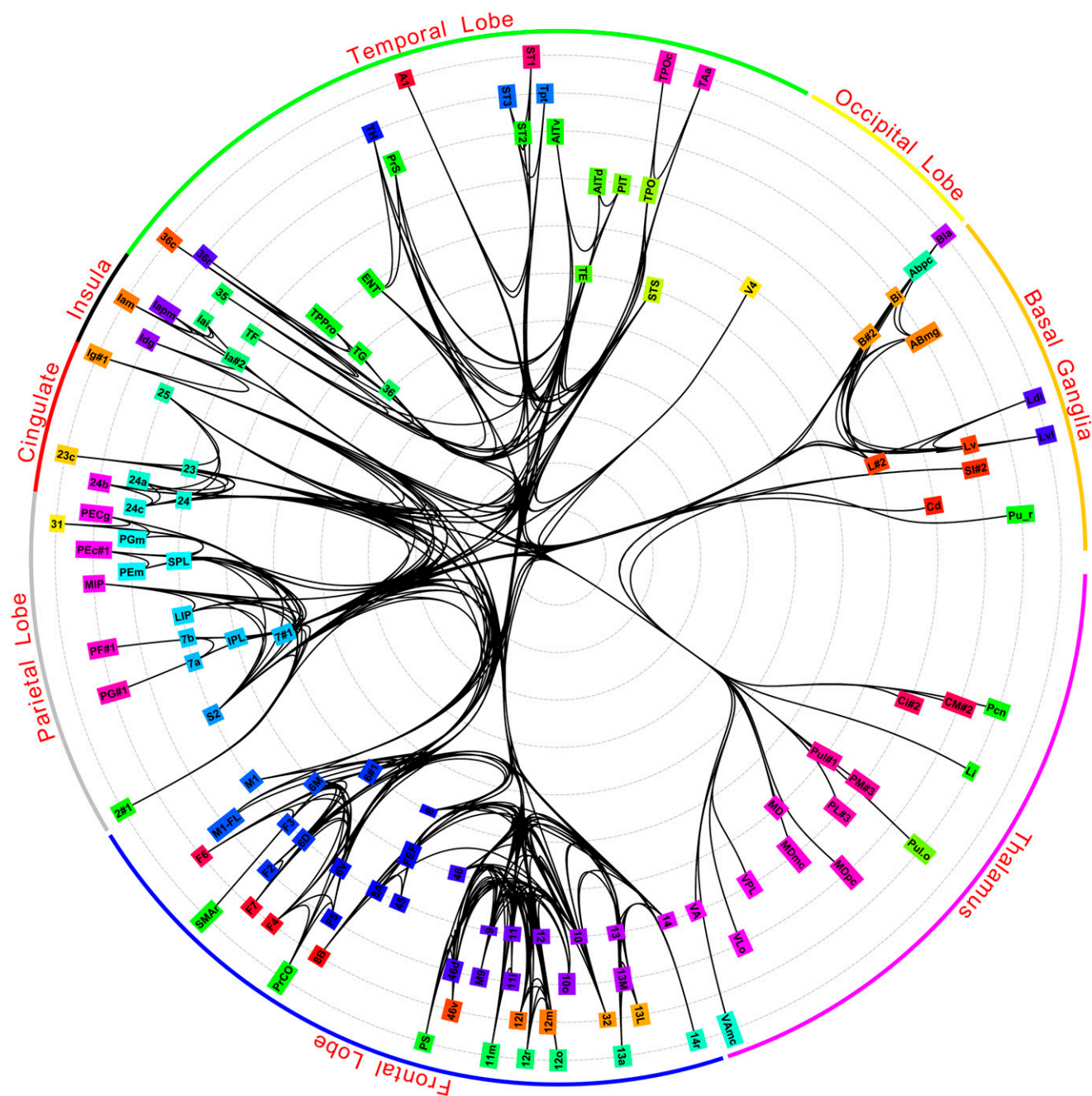


Fig. 3. Innermost core for the undirected version of our network. The innermost core is a central subnetwork that is far more tightly integrated than the overall network. Information likely spreads more swiftly within the innermost core than through the overall network, the overall network communicates with itself mainly through the innermost core, and the innermost core contains major components of the task-positive and task-negative networks derived via functional imaging research (43).

Further confusing the matter, Eguiluz et al. (33) found that functional networks of the human brain are scale-free, but Achard et al. (34) argued that at the level of resting state networks between cortical areas, these same networks are not scale-free. Restricted by the small size of available networks, Kaiser et al. (35) pursued an indirect approach based on simulated lesion studies (36) and concluded that “cortical networks are affected in ways similar to scale-free networks concerning the elimination of nodes or connections. However, a direct comparison of degree distributions has been impossible.”

Armed with our network, we provide a fresh perspective on the controversy. Based on the recipe for analyzing power law distributions in the study by Clauset et al. (37), Fig. 24 demonstrates

that the maximum likelihood scale-free hypothesis is untenable. Fig. 2B and *SI Appendix, Fig. S22* demonstrate that over the finite range of available data, the maximum entropy exponential distribution fits the data well. It is noteworthy that for the 302-neuron network in the worm *Caenorhabditis elegans* (38), the tail of the degree distribution is also well approximated by exponential decays (31).

Prefrontal Cortex Is Topologically Central. We have seen that vertices in our network have differing degrees of connectivity. We now introduce a number of widely studied metrics of topological centrality that take into account how vertices are interconnected.

Table 1. Top 10 brain regions according to several metrics of topological centrality for our network

Characteristic	Rank →	1	2	3	4	5	6	7	8	9	10
Integrator	In-degree	32	46	12o	12l	11	24	F7	14	8A	LIP
	In-closeness	46	12o	32, 11	24	12l	MD	8A	23c	8B	LIP, F7
	Authorities	32	12o	46	11	12l	24	14	F7	MD	9
Distributor	Out-degree	46	24	TF	9	13	13a	TH	TE, LIP	PGm	V2
	Out-closeness	46	24	TF	TE	9	TH	LIP	PGm	23, PM#3, 45	12
	Hubs	46	24	9	TF	TE	TH	13	32	23	PM#3
Intermediary	Betweenness	24	46	LIP	13a	MD	32	TF	PIT	13	PS
	PageRank	32	MD	46	36r	PIT	12o	24	23c	12l	11

The regions in prefrontal cortex are shown in bold. *SI Appendix, Table S1* provides an index of acronyms for the brain regions. The table was computed using Pajek (42).

In- and out-degrees, respectively, are direct measures of how much information a vertex receives and sends. For each vertex, define out-closeness as its average shortest path to every other vertex and its in-closeness as the average shortest path to it from every other vertex (40). For each vertex, define betweenness centrality as the number of shortest paths that pass through it (41, 40). PageRank was developed in the context of Web searching to find how often a vertex will be visited during random network traversal (18). Betweenness centrality and PageRank, which take both in- and out-connections into account, measure the efficacy of vertices in information intermediation. Hubs and authorities were also developed in the context of Web searching, and are defined relative to each other. They are recursively, circularly, and iteratively computed: A good hub links to many good authorities, and a good authority is one that is linked to by many good hubs (19). Hubs distribute information, whereas authorities aggregate information.

Table 1 shows the top 10 brain regions according to the above metrics of topological centrality. Roughly, 70% of the top 10 regions according to in-degree, in-closeness, and authorities reside primarily in prefrontal cortex (32, 46, 12o, 12l, 11, 14, 8A, 8B, 14, 9), suggesting that it serves as an integrator of information.

The top out-degree, out-closeness, and hub regions are distributed across prefrontal cortex (46, 9, 13, 13a, 45, 12, and 32), temporal lobe (TH, TF, and TE), parietal lobe (LIP and PGm), cingulate cortex (24 and 23), occipital lobe (V2), and thalamus (PM#3), with prefrontal cortex claiming 40% of the top 10 regions. This indicates that prefrontal cortex may also serve as a distributor of information. The top 10 regions according to betweenness and PageRank are distributed across prefrontal cortex (46, 13a, 32, 13, PS, 12o, 12l, and 11), temporal lobe (TF, PIT, and 36r), cingulate cortex (24 and 23c), parietal lobe (LIP), and thalamus (MD), with roughly half of the top regions residing in prefrontal cortex. Together, in a precise, quantitative, and multidimensional fashion, these facts strengthen the hypothesis that prefrontal cortex is an efficient intermediary of information serving both as an integrator and a distributor.

Is the topological centrality of prefrontal cortex an artifact of prefrontal regions being studied more often? Our investigation (*SI Appendix, Figs. S23–28*) did not find that prefrontal cortex (and its subregions) was studied more often than other brain regions in CoCoMac data, nor did it find a correlation between how often a region is studied and its degree. On the other hand, as expected, *SI Appendix, Fig. S29* finds that degree is correlated with centrality. Together, these facts imply that topological centrality of prefrontal cortex is not attributable to it being studied more often.

Anatomy Meets Physiology and Behavior. Topological centrality indicates that some vertices are more special than others. A logical ensuing question is whether the brain network contains special subnetworks. Now, we demonstrate that the brain network indeed contains a special subnetwork that captures its topological essence.

Core decomposition is a computationally efficient algorithm (17) that recursively peels off the least connected vertices to reveal progressively more closely connected subnetworks. In the first step, the algorithm recursively peels off all vertices with only

one edge until only vertices with at least two edges remain. In the second step, the algorithm recursively peels off all the vertices with only two edges until only vertices with at least three edges remain. The algorithm continues in like manner until all vertices are peeled off. Each peeling step defines a core. Each core is a subset of the previous core; hence, the cores constitute a nested hierarchy (*SI Appendix, Fig. S31*). Progressing along the hierarchy yields successive cores that are ever more tightly interconnected. The last or the innermost core is the top of this hierarchy and constitutes a topologically central subnetwork.

We found the innermost core for the undirected version of our network (Fig. 3), and it turned out to be a remarkable topological structure. The innermost core is deeply nested (*SI Appendix, Fig. S31*), such that each vertex in the innermost core touches at least 29 other vertices in the innermost core. The innermost core has 122 vertices. Let us refer to the set of remaining 261 vertices as the crust. There are 2,872 edges from the innermost core to itself, 1,707 edges from the crust to the innermost core, and 1,230 edges from the innermost core to the crust. There are only 793 edges from the crust to itself. Thus, 88% of all edges either originate or terminate in the innermost core, although it contains only 32% of the vertices. The longest shortest path (namely, diameter) for the innermost core is only 4, whereas for the overall network, it is significantly higher, namely, 6. Similarly, the average shortest path between any two vertices in the innermost core is only 1.95, whereas for the overall network, it is significantly higher, namely, 2.62. Further, the innermost core contains the vast majority of topological central vertices in Table 1 (*SI Appendix, Fig. S32*). Thus, the innermost core is a central subnetwork that is far more tightly integrated than the overall network, information likely spreads more swiftly within the innermost core than through the overall network, and the overall network communicates with itself mainly through the innermost core.

Although the innermost core is structurally interesting, it is functionally even more intriguing. The innermost core spans pre-motor and prefrontal cortex (42 regions), temporal lobe (23 regions), parietal lobe (16 regions), thalamus (15 regions), basal ganglia (12 regions), cingulate cortex (7 regions), insula (6 regions), and V4 in visual cortex. *SI Appendix* enumerates all brain regions in the innermost core. Three decades of functional brain imaging research in humans has culminated in the definition of two dynamically anticorrelated functional networks: a task-positive network activated during goal-directed performance and a task-negative network implicated in self-referential processing (43). Assuming a plausible set of homologies between human and macaque cortical organization,[§] we found that the innermost core contains major components of both of these networks (*SI Appendix*

[§]Establishing homology between human and macaque cortical organization remains an ongoing and active research area (23, 44–46), and it has been clearly noted that “homology cannot be proven but must be ‘inferred’” (47). Nonetheless, building on the conclusion in the article by Orban et al. (47) that “Despite several functional differences, many areas are homologous, especially at early levels of the visual hierarchy. In higher-order cortex, ‘regional’ homology still largely applies” and emboldened by the early functional MRI studies in mapping task-positive and task-negative networks in macaque (48), here, we assume that homology indeed holds.

and *SI Appendix, Fig. S33*). The innermost core constitutes the anatomical substrate that mediates temporally coordinated correlations within each network and anticorrelations between the networks and upholds physiological correlates underlying behavior.

Given the structural and functional centrality of the innermost core, it is natural to ask if it is sensitive to changes in the network. Quite reassuringly, precise analysis has revealed that the innermost core cannot change dramatically, given modest additions or deletion of edges in the network (*SI Appendix, Tables S5 and S6*); hence, it is an extremely stable and robust signature of the network.

Discussion

We have collated a comprehensive, consistent, concise, coherent, and colossal network spanning the entire brain and grounded in anatomical tracing studies that is a stepping stone to both fundamental and applied research in neuroscience and cognitive computing (14). What was previously scattered across 410 papers, 10,681 connectivity relations, and 16,712 mapping relations and limited to neuroanatomists specializing in the wetware of the experimental animals is now unified and accessible to network scientists who can unleash their algorithmic software toolkits (20–22).

We have begun to uncover remarkable global topological regularities of the network. The maximum entropy exponential distribution

$$(\text{average degree})^{-1} \exp[-x/(\text{average degree})]$$

characterizes the degree distribution of the network surprisingly well. Prefrontal cortex claims a disproportionately large share of topologically central brain regions according to a variety of ranking schemes, and thus serves as both an integrator and a distributor of information in the brain. We have found a deeply nested and tightly integrated core circuit spanning the entire brain that contains both the task-positive and task-negative networks. Assuming homology, it is indeed reassuring that the core circuit computed using structural data from a half century of anatomical tracing data in nonhuman primates corresponds so well with 3 decades of behavioral imaging research in humans. This hints at an evolutionarily preserved core circuit of the brain that may be a key to the age-old question of how the mind arises from the brain.

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